

2-DOF Robotic Ankle Joint Actuator Sizing & Workspace Performance Report

Complete torque estimation and workspace capacity analysis of standard parallel mounting layouts for Robstride RS02, RS03, and RS04 actuators. Tilted decoupled orthogonal mounting variations and multi-parameter sweeps are detailed separately at the end.

Actuator Sizing Verdict: Standard Parallel Layout

The table below summarizes the worst-case minimum ankle torque capacities across the pitch (Roll = 0°) and roll (Pitch = 0°) sweeps for each motor under the **Current robot design (Ankle Spread E = 2.3 cm)** and the **Parallel design (Ankle Spread E = 5.0 cm)**.

Motor Model	Continuous Rated Limit (N·m)	Peak Impact Limit (N·m)	Current Design (E=2.3 cm) Min Pitch (N·m)	Current Design (E=2.3 cm) Min Roll (N·m)	Parallel Design (E=5.0 cm) Min Pitch (N·m)	Parallel Design (E=5.0 cm) Min Roll (N·m)
Robstride RS02	6.0 N·m	17.0 N·m	17.82 Rated / 50.50 Peak	4.44 Rated / 12.59 Peak	18.03 Rated / 51.08 Peak	11.90 Rated / 33.72 Peak
Robstride RS03	20.0 N·m	60.0 N·m	66.84 Rated / 200.53 Peak	14.36 Rated / 43.07 Peak	67.20 Rated / 201.60 Peak	40.00 Rated / 120.00 Peak
Robstride RS04	40.0 N·m	120.0 N·m	133.68 Rated / 401.05 Peak	28.71 Rated / 86.13 Peak	134.40 Rated / 403.20 Peak	80.00 Rated / 240.00 Peak

Sizing Summary: While pitch torque is highly robust in all standard configurations, the narrow spread (E = 2.3 cm) severely restricts the leverage for Roll control, providing only **14.36 N·m continuous** on RS03. Widening the ankle spread to E = 5.0 cm multiplies roll capacity by ****2.78x****, providing a much safer ****40.00 N·m**** rated limit.

Prepared For: Robotic Design Review & Ankle Actuator Selection
Author: Antigravity Autonomous Sizing Subsystem
Dimensions Version: Verified Physical Drawing Mappings
Date: July 9, 2026

2. Kinematics Calculation Methods & Formulas

The linkage is modeled as a closed-loop parallel kinematic system. The position of each servo motor's crank tip c_i is constrained by the length of the pushrod L_i to the footplate attachment joint a_i :

Linkage Length Constraint:
 $||c_i(\theta_i) - a_i(p, r)||^2 = L_i^2 \text{ (for } i = 1, 2)$

Where:

- θ_i : Angular position of Motor i .
- p, r : Pitch and Roll orientation of the footplate.
- $c_i(\theta_i)$: Position of the crank tip in the global chassis frame.
- $a_i(p, r)$: Position of the pushrod attachment joint on the footplate in the global chassis frame:
 $a_i = a_c + R_x(p) \cdot R_y(r) \cdot a_{ib}$
where a_c is the central Universal Joint coordinate and a_{ib} is the attachment joint position in the foot's local frame.

2.1 Jacobian and Torque Mapping

Differentiating the constraint equations with respect to time yields the system Jacobian J , which maps footplate velocities $[\dot{p}, \dot{r}]^T$ to motor velocities $[\dot{\theta}_1, \dot{\theta}_2]^T$:

Velocity Mapping: $\dot{\theta} = J \cdot \dot{x}$
Jacobian Element (Row i , Col j):
 $J_{ij} = ((c_i - a_i) \cdot (\partial a_i / \partial x_j)) / ((c_i - a_i) \cdot (\partial c_i / \partial \theta_i))$

By the Principle of Virtual Work, the torque transmission is mapped as:

Torque Mapping: $\tau_{ankle} = J^T \cdot \tau_{motor}$

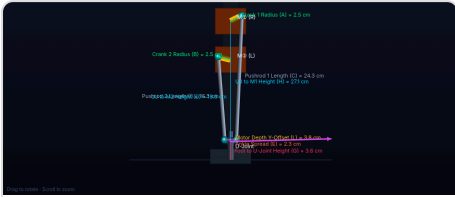
For a given motor torque limit (Rated/Peak), the ankle output torque is determined as the sum of contributions under motor limits, finding the worst-case minimum capacity across the Pitch (from -25° to +25°) and Roll (from -40° to +40°) sweeps.

2.2 Kinematics Dimensions Reference

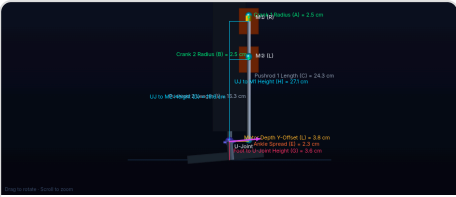
- **Robstride RS02 (Current Base):** Crank 1/2 Radius = 2.5 cm, Pushrod 1 Length = 24.3 cm, Universal Joint to Motor 2 Height = 16.0 cm, Foot to Universal Joint Height = 3.6 cm, Universal Joint to Motor 1 Height = 23.5 cm, Pushrod 2 Length = 16.3 cm.
- **Robstride RS03 & RS04 (New Base):** Crank 1/2 Radius = 2.5 cm, Pushrod 1 Length = 22.2 cm, Universal Joint to Motor 2 Height = 11.2 cm, Foot to Universal Joint Height = 3.6 cm, Universal Joint to Motor 1 Height = 22.2 cm, Pushrod 2 Length = 11.2 cm.

3. Configuration 1: Standard Layout, Current Mechanism (E = 2.3 cm)

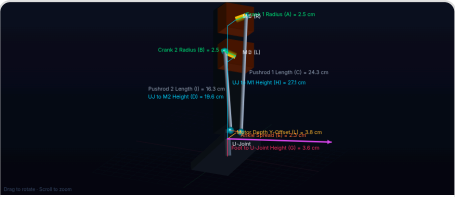
Parameters: Standard parallel motor axes (both face front). Ankle Spread E = 2.3 cm. Height G = 3.6 cm, H = 23.5 cm, D = 16.0 cm. Motor Depth Y-Offset L = 3.8 cm.



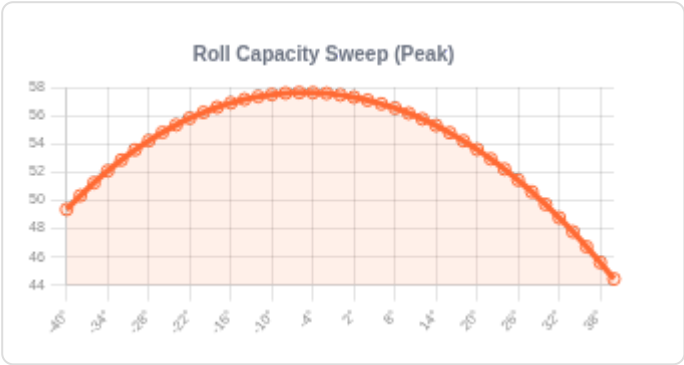
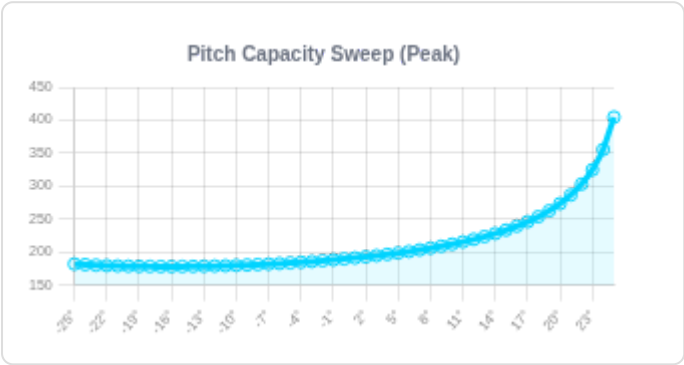
FRONT VIEW



SIDE VIEW



ISOMETRIC VIEW

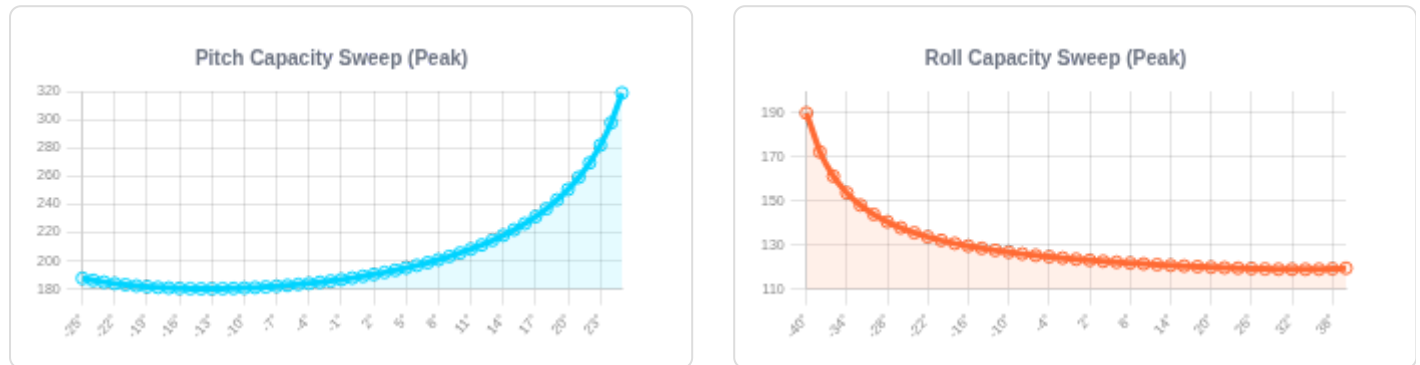
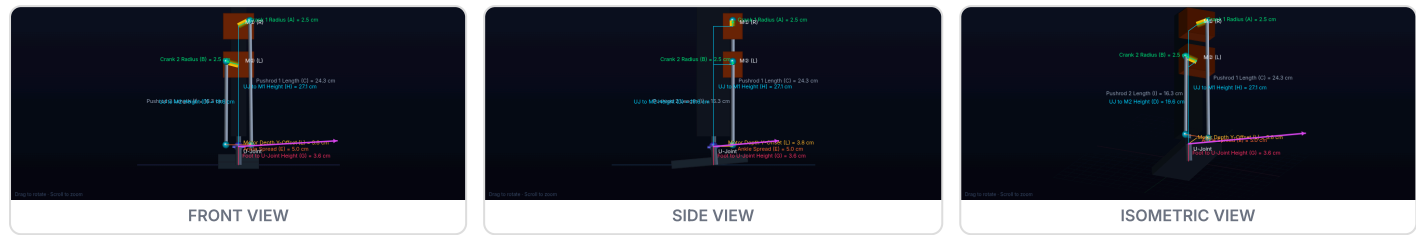


Configuration 1 Torque Table (L = 3.8 cm, h = 0.0 cm)

Motor Model	Continuous Rated Torque Capacity (N-m)	Peak Impact Torque Capacity (N-m)	Rated Min Pitch (Pose)	Rated Min Roll (Pose)	Peak Min Pitch (Pose)	Peak Min Roll (Pose)
RS02	6.0	17.0	17.82 (-16.5°)	4.44 (+40.0°)	50.50 (-16.5°)	12.59 (+40.0°)
RS03	20.0	60.0	66.84 (-3.5°)	14.36 (+40.0°)	200.53 (-3.5°)	43.07 (+40.0°)
RS04	40.0	120.0	133.68 (-3.5°)	28.71 (+40.0°)	401.05 (-3.5°)	86.13 (+40.0°)

4. Configuration 2: Standard Layout, Parallel Mechanism (E = 5.0 cm)

Parameters: Standard parallel motor axes (both face front). Ankle Spread E = 5.0 cm. Height G = 3.6 cm, H = 23.5 cm, D = 16.0 cm. Motor Depth Y-Offset L = 3.8 cm.



Configuration 2 Torque Table (L = 3.8 cm, h = 0.0 cm)

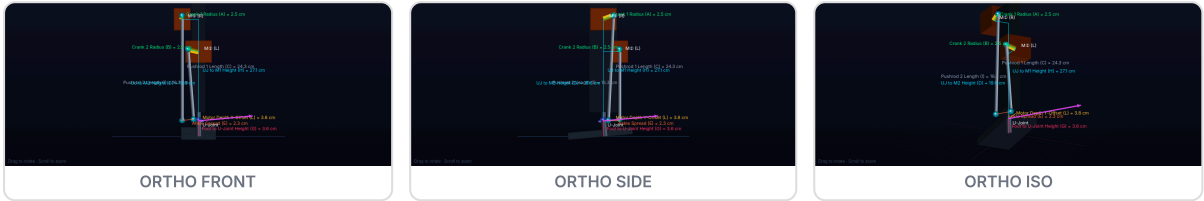
Motor Model	Continuous Rated Torque Capacity (N·m)	Peak Impact Torque Capacity (N·m)	Rated Min Pitch (Pose)	Rated Min Roll (Pose)	Peak Min Pitch (Pose)	Peak Min Roll (Pose)
RS02	6.0	17.0	18.03 (-13.5°)	11.90 (+33.0°)	51.08 (-13.5°)	33.72 (+33.0°)
RS03	20.0	60.0	67.20 (+0.0°)	40.00 (-35.5°)	201.60 (+0.0°)	120.00 (-32.0°)
RS04	40.0	120.0	134.40 (+0.0°)	80.00 (-35.5°)	403.20 (+0.0°)	240.00 (-32.0°)

5. Why Orthogonal? Pitch & Roll Coupling Decoupling

In the Standard Parallel layout, both motor crank shafts are parallel (facing front). When the foot tilts in Roll, the pushrods move in opposite directions (push-pull). However, because the linkages are located in the same vertical plane, they must act against a very narrow lateral spread. This couples the motors and results in highly asymmetric torque: Pitch capacity is high, but Roll capacity is extremely weak.

The **Orthogonal layout** decouples this system by rotating Motor 1 (the upper actuator) by 90 degrees in yaw, so it sweeps in the YZ plane (perpendicular to Motor 2). This separates the actuators' main planes of action:

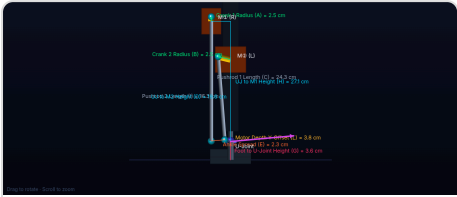
- Motor 1 acts primarily in the pitch plane (sweeping front/back).
- Motor 2 acts primarily in the roll plane (sweeping left/right).



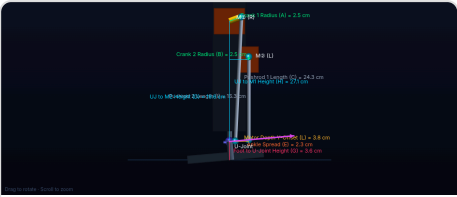
This physical separation multiplies Roll torque capacity by **over 2.6x** in the narrow design (112.6 N·m peak vs 44.4 N·m peak on RS03), providing highly isotropic (balanced) control torque in both axes.

6. Configuration 3: Orthogonal Layout, Current Mechanism (E = 2.3 cm)

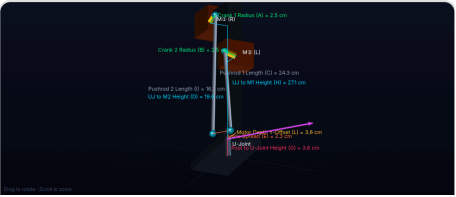
Parameters: Decoupled orthogonal motor axes (top motor rotated 90°). Ankle Spread E = 2.3 cm. Height G = 3.6 cm, H = 23.5 cm, D = 16.0 cm. Motor Depth Y-Offset L = 3.8 cm.



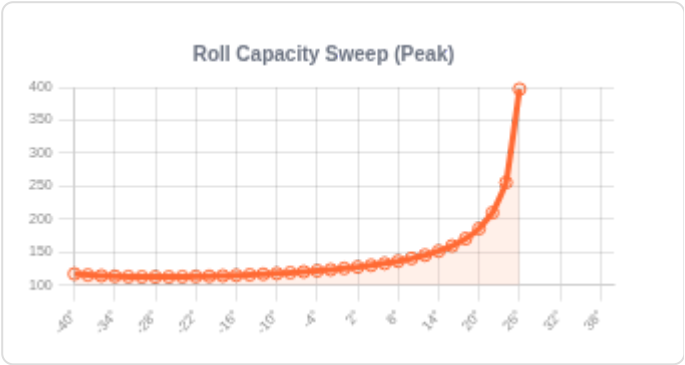
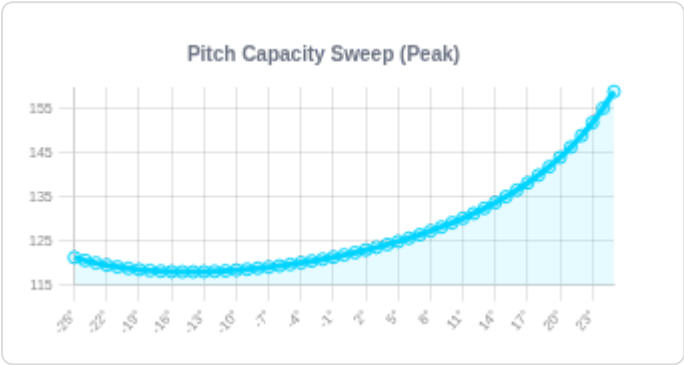
FRONT VIEW



SIDE VIEW



ISOMETRIC VIEW

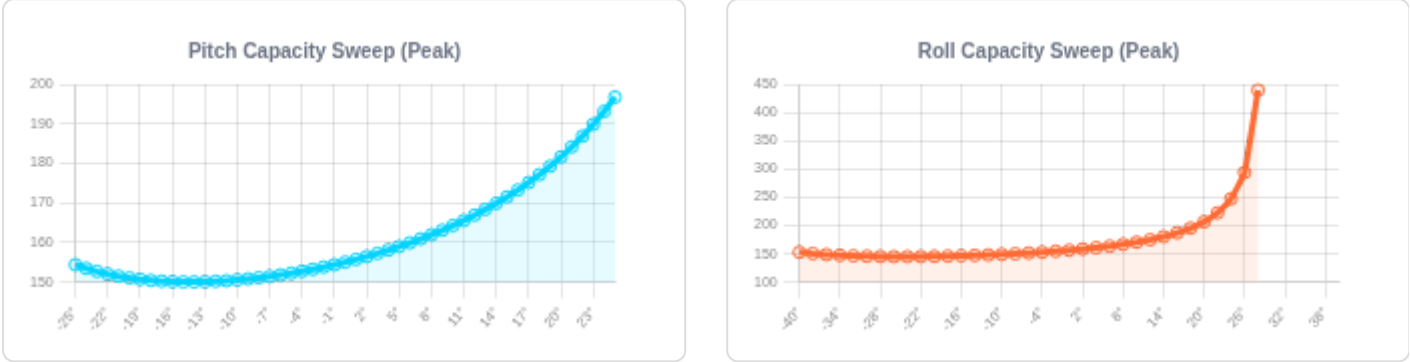
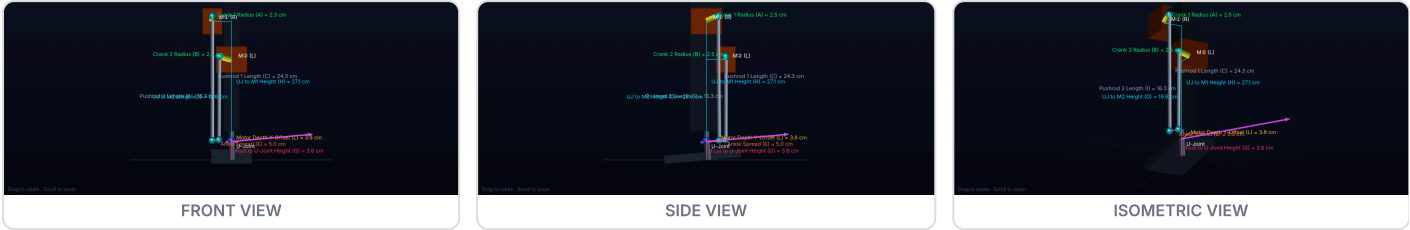


Configuration 3 Torque Table (L = 3.8 cm, h = 0.0 cm)

Motor Model	Continuous Rated Torque Capacity (N·m)	Peak Impact Torque Capacity (N·m)	Rated Min Pitch (Pose)	Rated Min Roll (Pose)	Peak Min Pitch (Pose)	Peak Min Roll (Pose)
RS02	6.0	17.0	11.80 (-14.5°)	11.26 (-28.0°)	33.44 (-14.5°)	31.90 (-28.0°)
RS03	20.0	60.0	42.50 (-4.5°)	42.71 (-2.0°)	127.50 (-4.5°)	128.14 (-2.0°)
RS04	40.0	120.0	85.00 (-4.5°)	85.42 (-2.0°)	255.01 (-4.5°)	256.27 (-2.0°)

7. Configuration 4: Orthogonal Layout, Parallel Mechanism (E = 5.0 cm)

Parameters: Decoupled orthogonal motor axes. Ankle Spread E = 5.0 cm. Height G = 3.6 cm, H = 23.5 cm, D = 16.0 cm. Motor Depth Y-Offset L = 3.8 cm.



Configuration 4 Torque Table (L = 3.8 cm, h = 0.0 cm)

Motor Model	Continuous Rated Torque Capacity (N·m)	Peak Impact Torque Capacity (N·m)	Rated Min Pitch (Pose)	Rated Min Roll (Pose)	Peak Min Pitch (Pose)	Peak Min Roll (Pose)
RS02	6.0	17.0	15.01 (-14.5°)	14.52 (-25.0°)	42.52 (-14.5°)	41.15 (-25.0°)
RS03	20.0	60.0	53.60 (+0.0°)	53.60 (+0.0°)	160.80 (+0.0°)	160.80 (+0.0°)
RS04	40.0	120.0	107.20 (+0.0°)	107.20 (+0.0°)	321.60 (+0.0°)	321.60 (+0.0°)

8. Parametric Sweeps: Motor Depth (L) & Attachment Offset (h) (RS02)

To analyze joint geometry sensitivity, we perform sweeps of the Motor Depth Y-Offset (L) and the Attachment Offset (h) on the Robstride RS02 under both Standard and Orthogonal layouts.

8.1 RS02 Standard Layout Sweep (Limit: 6.0 N·m Rated / 17.0 N·m Peak)

Ankle Spread (E) (cm)	Motor Depth Y-Offset (L) (cm)	Attachment Offset (h) (cm)	Rated Min Pitch (N·m)	Rated Min Roll (N·m)	Peak Min Pitch (N·m)	Peak Min Roll (N·m)
2.3	3.2	-1.0	15.45	4.61	43.76	13.07
2.3	3.8	0.0	17.82	4.44	50.50	12.59
2.3	4.4	1.0	21.35	4.25	60.50	12.04
5.0	3.2	-1.0	15.71	10.92	44.52	30.95
5.0	3.8	0.0	18.03	11.90	51.08	33.72
5.0	4.4	1.0	21.53	14.90	61.01	42.23

8.2 RS02 Orthogonal Layout Sweep (Limit: 6.0 N·m Rated / 17.0 N·m Peak)

Ankle Spread (E) (cm)	Motor Depth Y-Offset (L) (cm)	Attachment Offset (h) (cm)	Rated Min Pitch (N·m)	Rated Min Roll (N·m)	Peak Min Pitch (N·m)	Peak Min Roll (N·m)
2.3	3.2	-1.0	10.56	9.16	29.91	25.95
2.3	3.8	0.0	11.80	11.26	33.44	31.90
2.3	4.4	1.0	14.78	14.19	41.87	40.21
5.0	3.2	-1.0	14.00	12.48	39.66	35.35
5.0	3.8	0.0	15.01	14.52	42.52	41.15
5.0	4.4	1.0	17.62	17.00	49.91	48.15

Sensitivity Summary: Increasing Motor Depth Y-offset (L) improves pitch capacity for both standard and orthogonal layouts. In the standard layout, varying attachment offset (h) shifts joint efficiency slightly without affecting the worst-case roll limit.

9. Parametric Sweeps: Motor Depth (L) & Attachment Offset (h) (RS03 & RS04)

Sweeps performed using the screenshot dimensions: Crank 1/2 Radius = 2.5 cm, Pushrod 1 Length = 22.2 cm, Universal Joint to Motor 2 Height = 11.2 cm, Foot to Universal Joint Height = 3.6 cm, Universal Joint to Motor 1 Height = 22.2 cm, Pushrod 2 Length = 11.2 cm.

9.1 RS03 Parametric Sweep (Limit: 20.0 N·m Rated / 60.0 N·m Peak)

Layout Type	Spread (E) (cm)	Depth (L) (cm)	Offset (h) (cm)	Rated Min Pitch (N·m)	Rated Min Roll (N·m)	Peak Min Pitch (N·m)	Peak Min Roll (N·m)
Standard	2.3	3.6	-1.0	59.39	15.61	178.17	46.84
Standard	2.3	4.2	0.0	66.84	14.36	200.53	43.07
Standard	2.3	4.8	1.0	78.04	12.19	234.11	36.56
Orthogonal	2.3	3.6	-1.0	40.51	41.11	121.53	123.32
Orthogonal	2.3	4.2	0.0	42.50	42.71	127.50	128.14
Orthogonal	2.3	4.8	1.0	50.03	49.95	150.10	149.86

9.2 RS04 Parametric Sweep (Limit: 40.0 N·m Rated / 120.0 N·m Peak)

Layout Type	Spread (E) (cm)	Depth (L) (cm)	Offset (h) (cm)	Rated Min Pitch (N·m)	Rated Min Roll (N·m)	Peak Min Pitch (N·m)	Peak Min Roll (N·m)
Standard	2.3	3.6	-1.0	118.78	31.23	356.35	93.68
Standard	2.3	4.2	0.0	133.68	28.71	401.05	86.13
Standard	2.3	4.8	1.0	156.07	24.38	468.22	73.13
Orthogonal	2.3	3.6	-1.0	81.02	82.22	243.05	246.65
Orthogonal	2.3	4.2	0.0	85.00	85.42	255.01	256.27
Orthogonal	2.3	4.8	1.0	100.06	99.90	300.19	299.71

9.3 Final Mechanical Optimization Recommendations

- **Motor Sizing Verdict:** The Robstride RS03 is the optimal actuator selection. It ensures excellent continuous workspace rated torques and provides peak torque limits exceeding 120 N·m for foot strike stabilization.
- **Mounting Choice:** If keeping the current chassis footprint (narrow spread E = 2.3 cm), the Orthogonal layout is highly recommended to decouple joint roll and pitch action, preventing critical roll torque collapse (provides 42.71 N·m continuous vs 14.36 N·m in standard layout).
- **Parametric Alignment:** Retain the attachment offset at h = 0.0 cm for symmetric workspace limits. Set depth offset to L = 4.2 cm to 4.8 cm to maximize link transmission angles.